NIMT - CALIBRATION NETWORK LABORATORIES

NIMT-CNL-SP No. 401

# Standard Procedure for Calibration of Industrial Platinum Resistance Thermometers Type/Model : Pt100

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Approved by NIMT

NIMT

# INDUSTRIAL PLATINUM RESISTANCE THERMOMETERS

## 1. Calibration Description:

| | | | | | | |
|---|---|---|---|---|---|---|
| **Type/Model** | **:** | **PT100** | | | | |
| **Temperature Range** | **:** | **- 40 °C to + 550 °C** | | | | |
| **Best Measurement Capability** | **:** | **0.03 °C** | **from** | **– 40°C** | **to** | **0 °C** |
| | | **0.03 °C** | **from** | **> 0 °C** | **to** | **250 °C** |
| | | **0.03 °C** | **from** | **> 250°C** | **to** | **420 °C** |
| | | **0.05 °C** | **from** | **> 420°C** | **to** | **550 °C** |

## 2. Scope

2.1 These methods describe the principles, necessary apparatus, and procedures for calibration and testing of platinum resistance thermometer ( PRTs) or industrial resistance thermometers (IPRTs).

2.2 These methods cover only the tests for insulation resistance, calibration, and thermal cycling stability at ice point (hysteresis).

2.3 These methods are not necessarily intended for, recommended to be performed on, or appropriate for every type of thermometer.

2.4 Thermometer performance minimum specifications and acceptance limits are included.

2.5 The standard specifies requirements for Industrial Platinum Resistance Thermometer sensor, PRT / IPRT. Temperature range – 40 °C to 550 °C

## 3. Reference Standard Specification Codes

The temperature / resistance relation ships used in this procedure are base on the EN DIN 60751 : 1995

## 4. Purpose

**To define the relation ship between resistance and temperature:**

*** For the range – 40 °C to 0 °C the interpolation equation is:**

$$R_t = R_0\,[1 + At + Bt^2 + Ct^3(t-100)]$$

*** For the range 0 °C to 550 °C the interpolation equation is:**

$$R_t = R_0\,[1 + At + Bt^2\,]$$

**And by Using the Callendar – Van Dusen equation, the calculated values of the constants *A*, *B* and *C* for the quality of commonly Industrial Platinum Resistance Thermometer in these equation can be obtained.**

**This paper presents the methods, equipment, and uncertainties associated with the calibration of industrial thermometers in the Temperature Laboratory.**

**The form and complexity of the calibration procedure will depend on both the accuracy and the temperature range required.**

## 5. Tolerances

The minimum requirement of temperature tolerance allowed in EN DIN 60751 at temperature, $t$ (°C), for normal two classes of IPRT's are as follows:

Class A : $\pm(\ 0.15 + 0.002\ |t|\ )$ and
Class B : $\pm(\ 0.3 + 0.005\ |t|\ )$

# 6. Equipment Require

| Description | Specification | Model/Type |
|---|---|---|
| 1. Apparatus | Usable Range : – 40°C to 550°C | See 6.1 for details |
| | Uniformity : ≤ 0.03°C or depend on capability | |
| | Stability : ≤ 0.03°C or depend on capability | |
| 2. Reference Thermometer | Usable Range : – 40°C to 660°C | See 6.2 for details |
| | Uncertainty : ≤ 0.02°C or depend on capability | |
| 3 Measurement Instruments | Usable Range : – 40°C to 660°C | See 6.3 for details |
| | Uncertainty : depend on capability | |

## 6.1 Apparatus

The necessary but not limited to apparatuses for this calibration method are:

6.1.1 Ice-Point Bath ; The simplest fixed point with an error of less than 0.01 °C

6.1.2 Triple Point of Water ; To accurately realize the triple point of water, a triple point of water cell is used.

6.1.3 Fluid Baths ; For controlling the temperature of various type of fluid baths by adjusting the amount of heating or cooling.
Water Baths, Salt Baths or Refrigerated Baths

**Table 1.** Fluid Bath Media and Typical Operating Temperature Range

| Media | Estimated Temperature Range (°C) |
|---|---|
| Silicone Oils | - 100 to 315 |
| Light Mineral Oils | - 75 to 200 |
| Alcohol | - 100 to 5 |
| Water | 0.5 to 100 |
| Molten Salts | 200 to 620 |
| Dry Fluids(Fluidized Particle) | 75 to 850 |

6.1.4 Temperature Furnaces ; For controlling the temperature by adjusting the amount of heating

6.1.5 Block Calibrations ; For controlling the high temperature by adjusting the amount of heating

### 6.2 Reference Thermometers

Standard Thermometer ; SPRT, PRT, T/C etc. to use with Digital Indicators

### 6.3 Measurement Instruments

There are several types of measurements can be used to monitor the measurands.

- AC or DC Bridges ; These instruments typically provide the option of a digital display which can be set to provide readings in ohms, or temperature.
- Digital Indicators ; Digital Thermometer or Digital Multi-meter

Note : All Fluid Baths, Temperature Furnaces and Block Calibrators must be evaluated to ensure the qualification of them be stable with time and uniform over the working space at the measuring temperatures.

## 7. Facilities or Controlled conditions

Laboratory Temperature Condition (23 ± ?) °C,

Relative Humidity (50 ± ?) % RH

Air ventilation system for harmful at high temperature operation.

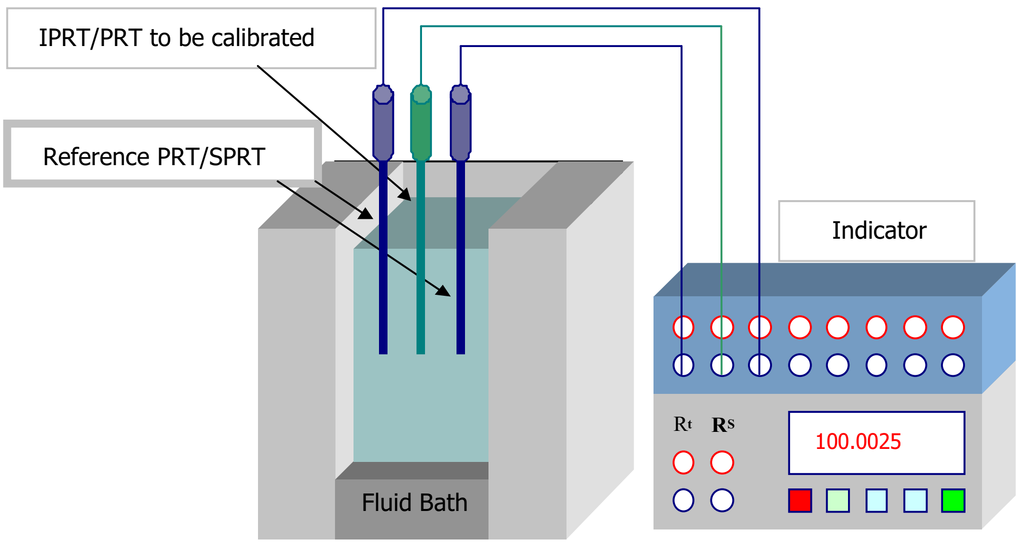
# 8. Measurements to be carried out

## 8.1 Preliminary operation

- Visual inspection (any damages, clean contacts, etc.)
- Initial check for connection wire configuration (2, 3 or 4 wires)
- Insulation check (resistance between sheath and leads must be higher than 1 G Ω)
- Ice-point or triple point check at appropriate current (initial measurement at the ice-point or water-triple point)

## 8.2 Preconditioning

- The IPRTs which require to calibrate exceed 400 °C should be annealing at 450 °C at lease two hours before proceeding of measurements.
- Then measure the IPRT's resistance value at ice point or WTP , the difference between before and after annealing should not greater than 6 m Ω which corresponding to temperature about 15 mK.
- Prepare calibration system as figure below ; (one or two SPRTs can be selected and assigned as reference standards)




- Carry out the measurements according to the step 8.3 and complete the result sheets.

### 8.3 Calibration Procedures

Before performing the measurement, the test IPRT must be undergone preconditioning procedure and then allow the temperature to ambient temperature.

- During the calibration process, thermometer is required to immerse at least 10 times of thermometer sheath's diameter plus sensor length below the liquid surface.(normally the immersion depth should be 150 mm to 200 mm)
- The immersion depth and measurement results shall be recorded in the result sheets.
- The IPRTs can be calibrated either from the lowest to the highest calibration point of the range or vice versa.
- The suggestion is it should be started from maximum temperature to the minimum temperature.

8.3.1 Before starting the first calibration point, set the temperature at 0 °C (or 0.01°C).

8.3.2 Wait until the reference standard reading stabilized then record readings on the calibration worksheet.

Record the ice point (0°C) or WTP readings before and after measurement cycle. The difference in ice-point reading is hysteresis (thermal cycling short term stability) in the uncertainty budget.

For normal IPRT, apply the proper sensing current as required by customer, if no specific current the sensing current of 1 mA should be applied for 5 to 10 set of measurements, the following measurement sequence at least 10 min. and record should be followed:

$$\textit{Std.} \rightarrow \textit{UUC} \rightarrow \textit{Std.} \rightarrow \textit{UUC} \rightarrow \textit{Std.} \rightarrow \ldots \rightarrow \textit{Std.}$$

[*For High Quality PRT, apply the sensing current of 1 mA for 10 set of measurements and then apply 1 x* $\sqrt{2}$ *mA for 10 set of measurements for self heating correction, follow measurement sequence at least 10 min and record the data:*]

8.3.3 Move all thermometers to ambient temperature and then put them in the desired fluid bath or furnace depend on required point.

8.3.4 Adjust the temperature of the liquid bath or furnace to the required calibration point and wait for stabilize. (normally starts the first point from the highest to the lowest temperature)

8.3.5 Repeat sequence of measurements after stabilize.

8.3.6 Set the bath controller (or temperature furnace) to the next lower required calibration points, wait until stabilize then record sequence of measurements.

8.3.7 Repeat 7.3.4 to 7.3.6 to complete all required calibration points.

8.3.8 Repeat measurement at ice point ( 0 °C) or WTP again to check thermal cycling stability and to complete the calibration process.

*If the thermal cycling stability is higher than the expected uncertainty, perform the annealing process at 450°C by more than 8 hours and perform the process of measurement again.*

8.3.9 Disconnect all equipment. End of measurement process

8.3.10 Calculate the results and relate uncertainties.

- Calculation of the temperature according to DIN EN 60751
- Calculation of the deviation function to DIN EN 60751
- Critical analysis of the deviation function
- Calculation of uncertainty

8.3.11 Report the calibration results.

This calibration guideline provides calibrations of industrial thermometers over the range from − 40 °C to 550 °C. The different types of thermometers include industrial platinum resistance thermometers (IPRTs) and platinum resistance thermometer ( PRTs) can be performed by both comparison with an ITS-90 calibrated standard platinum resistance thermometer (SPRT) calibrated on the International Temperature Scale of 1990 and by a limited number of fixed-points.

The comparison baths are liquid baths alcohol ( − 80°C to 1°C), water (0.5°C to100 °C), oil (50°C to 250°C), and salt (150°C to 550°C).

The fixed points may be used are the melting point of ice (0 °C), the triple point of water (0.01 °C), and the melting point of gallium (29.7646 °C).

The temperature measurement system for the reference SPRT it is a commercially-available digital thermometer and/or ac resistance ratio bridge, for the IPRTs.

All of these temperature measurement systems are integrated via the computer data acquisition systems that semi-automate calibrations. This paper presents the methods, equipment, and uncertainties associated with the calibration of industrial thermometers in the Temperature Laboratory.

The calibration report provides an ITS-90 based table of measured bath or fixed-point temperature versus thermometer resistance, calibration uncertainties, and the thermometer calibration coefficients.

The IPRT/ PRT calibrations are performed by following the instruction on the EN DIN 60751.

The typical IPRT's nominal resistance values at 0 °C, $R_0$, are normally 100 Ω

The IPRT calibrations results are present in the form of the interpolation equations defined in ITS-90.

To define the relation ship between resistance and temperature:

* For the range – 40 °C to 0 °C the interpolation equation is:

$$R_t = R_0\,[1 + At + Bt^2 + Ct^3(t-100)]$$

* For the range 0 °C to 550 °C the interpolation equation is:

$$R_t = R_0\,[1 + At + Bt^2\,]$$

And by Using the Callendar – Van Dusen equation, the calculated values of the constants for the quality of commonly Industrial Platinum Resistance Thermometer in these equation are:

$A$ = $3.9083 \times 10^{-3}$ °C$^{-1}$

$B$ = $-\,5.775 \times 10^{-7}$ °C$^{-2}$ [the constant $B$ should be about $-\,5.75 \times 10^{-7}$5 to $-\,5.89 \times 10^{-7}$ ]

$C$ = $-\,4.183 \times 10^{-12}$ °C$^{-4}$

For the IPRT satisfying the above relationships, the average slope of the ratio W-versus-t between 0 °C and 100 °C or the temperature coefficient, defined as:

$$\alpha = \frac{R_{100} - R_0}{100 \times R_0}$$

**Where:**

$R_0$ is the resistance at 0 °C;

$R_{100}$ is the resistance at 100 °C;

**The $\alpha$ is used as the PRT sensors purity criteria:**

$\alpha$ = 0.00385 °C$^{-1}$ for sensor complying with EN DIN-60751

$\alpha$ ≈ 0.003920 °C$^{-1}$ for many secondary standard PRT's.

**Tolerances**

The minimum requirement of temperature tolerance allowed in EN DIN 60751 at temperature, $t$ (°C), for normal two classes of IPRT's are as follows:

Class A : $\pm\,(0.15 + 0.002\,|t|\,)$ and

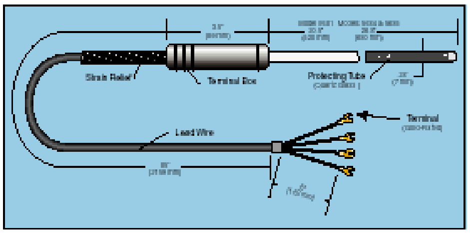
Class B : $\pm\,(0.3 + 0.005\,|t|\,)$

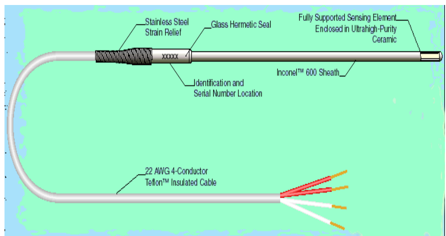
| Temperature (°C ) | TOLERANCE | | | |
|---|---|---|---|---|
| | Class A | | Class B | |
| | (±°C ) | (±Ω) | (± °C ) | (±Ω) |
| - 200 | 0.55 | 0.24 | 1.3 | 0.56 |
| - 100 | 0.35 | 0.14 | 0.8 | 0.32 |
| 0.000 | 0.15 | 0.06 | 0.3 | 0.12 |
| 100 | 0.35 | 0.13 | 0.8 | 0.30 |
| 200 | 0.55 | 0.20 | 1.3 | 0.48 |
| 300 | 0.75 | 0.27 | 1.8 | 0.64 |
| 400 | 0.95 | 0.33 | 2.3 | 0.79 |
| 500 | 1.15 | 0.38 | 2.8 | 0.93 |
| 600 | 1.35 | 0.43 | 3.3 | 1.06 |
| 650 | 1.45 | 0.46 | 3.6 | 1.13 |
| 700 | | | 3.8 | 1.17 |

**Thermometer Connecting Wire Configuration Check**

Connecting wire Configuration

Identification marking or color coding




Check that the correct wires are connected to the properly terminal of digital thermometer or the resistance bridge as out line above. In the even of unclear marked wires, the following procedure should be performed.

1.1 The resistance of sensing wires across the sensing element should be approximately the $R_0$. (about 100Ω for Pt 100 IPRT etc.)

1.2 The resistance of current wires across the sensing element should be approximately the $R_0$. (about 100Ω for Pt 100 IPRT etc.)

1.3 The resistance of the connecting wires at the same side of the sensing element should be approximately zero.

1.4 Measure the cable length for 2 wires or 3 wires connections.

Total uncertainty for comparison calibration of thermometers consists always of the same general parts:

| Uncertainty contribution of the standard | Uncertainty contribution of the furnace/bath | Uncertainty contribution of the UUT | Uncertainty contribution of the approximation/ interpolation function |
|---|---|---|---|
| $\delta t_S$ | $\delta t_F$ | $\delta t_{m,X}$ | $\delta t_{Func}$ |

## General Model Function:

$$\Delta t_X = t_S - t_X + \delta t_S + \delta t_F + \delta t_{m,X} + \delta t_{Func}$$

### Uncertainty of the calibration

$$u(\Delta t_X) = \sqrt{u^2(t_S) + u^2(\delta t_F) + u^2(\delta t_{m,X}) + u^2(\delta t_{Func})}$$

### Uncertainty contribution of the standard

The uncertainty for the standard can be estimated by a sub-budget:

$$\delta t_S = \delta t_{Cal\_S} + \delta t_{Drift\_S} + \delta t_{Cal\_M} + \delta t_{Drift\_M}$$

with:

| | |
|---|---|
| $\delta t_{Cal\_S}; \delta t_{Cal\_M}$ | Calibration uncertainty of thermometer (sensor) and electrical measuring device (expanded uncertainty from last recalibration) |
| $\delta t_{Drift\_S}; \delta t_{Drift\_M}$ | Uncertainty of thermometer (sensor) and electrical measuring device due to the drift since last recalibration (typical rectangular distribution.) |

**Uncertainty Contribution of the furnace/bath**

The uncertainty for the furnace/bath can be estimated by a sub-budget:

$$\delta t_F = \delta t_{Inhom} + \delta t_{Instab}$$

with:

$\delta t_{Inhom}$ Local inhomogeneity of the temperature distribution inside the calibration volume of the furnace, block or bath (typical rectangular distribution.)

$\delta t_{Insta}$ Instability of the temperature inside the calibration volume of the furnace, block or bath (typical rectangular distribution.)

*The information regarding inhomogeneity and instability can be obtained from periodical investigations of the thermal properties of the furnace or bath.*
*Specifications of the manufacturer can be taken into account but have to be verified.*

**Uncertainty Contribution of the UUT**

The uncertainty for UUT can be estimated together with measurement instrument by a sub- budget:

$$\delta t_{m,X} = \delta t_{UUC} + \delta t_{hyst} + \delta t_{Lin} + \delta t_{res} + \delta t_{Cal\_M} + \delta t_{Driftl\_M} + \delta t_{Lin\_M}$$

**with:**

Uncertainty due to UUC

$\delta t_{UUC}$ : Type A uncertainty of the thermometer reading (standard deviation of the reading-normal distributed k=1)

$\delta t_{hyst}$ : Uncertainty contributions of thermometer under test (sensor) due to thermal cycling short term stability at ice point is assume as hysteresis. (typical rectangular distributed)

Uncertainty due to Indicator

$\delta t_{res}$; $\delta t_{Lin}$ : Uncertainty contributions of thermometer under test (sensor) due to resolution limits, nonlinearity, …

(typical rectangular distributed)

$\delta t_{Cal\_M}$; $\delta t_{Drift\_M}$ : Uncertainty contributions of the electrical measuring device used for UUT measurement due to calibration, drift, …

(typical rectangular or normal distribution)

**The measurement**

The Industrial Platinum Resistance Thermometer (IPRT) which nominal value about 100 Ω at 0 °C, had been calibrated by comparing with Standard Platinum Resistance Thermometer and a high precision calibrated digital thermometer.

The resistance of the IPRT being calibrated was measured using the 4 - terminal input of the high precision digital thermometer.

The two probes were immersed at approximately 200 mm immersion depth and compared in stirred liquid baths (water and oil) over the range 0 °C to 250 °C.

Measurements at 0 °C were done only at the start and finish of the sequence.

**The model**

The aim in calibration is to obtain the correction $C$:

$$C = \Delta t_X = t_S - t_X; \qquad (1)$$

where $t_S$ is the `true' temperature derived from the calibrated reference sensor (corrected for its calibration) and $t_X$ is the value of temperature given for the IPRT by:

$$t = a(w-1) + b(w-1)^2 + c(w-1)^3 ; \qquad (2)$$

Where:

$w = R_t / R_0$

$R_t$ and $R_0$ are the resistances of the IPRT at temperature $t$ and at 0 °C, respectively.

The minimum uncertainty components should be considered are described below:

**Reference temperature** (Uncertainty contribution of the standard)

1. Calibration of reference thermometer:

   Standard thermometer (SPRTs or PRTs) was calibrated with reported uncertainty of ± 6 mK (Ui = 6 mK) over the range 0 °C to 420 °C.

   The stated coverage factor was 2 (corresponding to $\nu_i = \infty$. : table of Student's values ).

2. Use of thermometer (Thermometer's drift):

   The drift in calibration was the only contribution that needed consideration. In this example, an uncertainty of ± 2 mK, based on history of the thermometer is chosen.

   Moreover, drift is thought more likely to be linear (rectangular distribution) so, $U_i$ was taken as the half-width 2 mK, half the extent of possible drift for the period, and the reducing factor *ki* as 1.732. Being a reasonable estimate, $\nu_i = \infty$.

3. Calibration of standard digital thermometer:

   The calibration uncertainty corresponding to term of temperature was ± 1 mK (*Ui* = 1 mK) over the range 0 °C to 420 °C and the calibration was done at an ambient temperature of 23 ± 2 °C.

   The stated coverage factor was 2 (corresponding to $\nu_i = \infty$. : table of Student's values ).

4. Drift of standard digital thermometer:

   The drift of digital thermometer over period of using corresponding to temperature was 4 mK and the reasonable uncertainty ( $U_i$ = 4 mK) is considered rectangular with $k_i$ = 1.732 and $\nu_i = \infty$.

5. Type A uncertainty in mean reading :

   For example at 100° C the observed range (*R*) of data displayed on the digital thermometer was 0.1 mK over (nominally) 30 readings

   The uncertainty calculated as $U_i$ = 0.1 mK (for $n \sim 30$, $k_i$ = 2 and $\nu_i = \infty$.).

**The IPRT**

6. Stability of IPRT at 0 °C:

   Ice point measurements of the IPRT at the start and finish of the calibration sequence differed by 0.4 m$\Omega$ which corresponding to temperature of about 1 mK.

   Consider this value as a reflection of short-term instability in the sensor and assume it was due to linear drift. Its effect was assumed rectangular with $U_i$ = 0.5 mK, $k_i$ = 1.732and $\nu_i = \infty$ .

7. Type A uncertainty in mean reading :

   The standard deviation of UUC resistance measurement at 100 °C observed on the digital thermometer can be translated to the values of temperature 0.1 mK over (nominally) 30 readings

   The uncertainty calculated translate to the values as $U_i$ = 0.1 mK (for $n$ ~30, $k_i$ = 2 and $\nu_i = \infty$.).

8. Calibration of Digital Thermometer ( or DMM)

   The 1 k$\Omega$ range of the Digital Thermometer or DMM had been calibrated at an ambient temperature of 23 ± 2°C.

   The calibration uncertainty was ± 3 ppm , based on a coverage factor of 2.

   The DMM was used for the measurements $R_t$ at 100°C, $R_t$ = 139$\Omega$, and the corresponding expanded uncertainties for the calibration are 0.4m $\Omega$ which corresponding to ≈ 1 mK.

9. Drift of Digital Thermometer ( or DMM)

   The Digital Thermometer or DMM was used under the same conditions (ambient temperature, integration time, etc) that applied during its calibration, so drift was the only relevant component. Such drift would already have been included in item 8, above.

10. Resolution of Digital Thermometer (DMM)

    The resolution uncertainty is ± half the least count of 0.1 m$\Omega$. (corresponding to 0.26 mK)

    It would affect both resistance measurements, in the sense that it represents the lower limit for uncertainty in their measurement. So, apply it once and use the largest sensitivity coefficient, that for Ro. The uncertainty would have a rectangular distribution with ki = 1.732.

    In this example this component have been already included in item 7, above.

11. Linearity of Digital Thermometer (DMM)

    This uncertainty component is also included in the uncertainty of calibration.

**Fluid Bath and Furnace**

12. Temperature difference (both uniformity and instability of baths and furnace): assume the spatial variation in the oil bath at 100 °C was assessed as 10 mK over the region in which PRT sensors are generally placed for calibration, being the maximum difference between any two temperatures measured in the region. It was assumed from this that there was a 95 % chance that the temperature difference between any two probes was within the range ± 10 mK, i.e., *Ui* = 10 mK.

**Interpolation Function Fitting**

13. Fitting and interpolation: calibrations were done at six temperatures, from which six values of *C* were obtained, scattering about zero. The approximately range of *C* values at 100 °C was 20 mK, so we have *Ui* = 10 mK. This selects the goodness of fit for the chosen cubic.

Uncertainty calculation Using the above values, the calculation of expanded uncertainty proceeds as follows:

the combined standard uncertainty, $u_c$ = 9,1 mK

the number of effective degrees of freedom, $\nu_{eff} \geq 200$

the coverage factor, k = 2,0

the expanded uncertainty, U = 18,2 mK.

so after rounding of expanded uncertainty, U = 18 mK.

**RESULTS:**

The measurement results of the IPRT are reported in the table below:

| Temperature (*t*) (°C) | UUC Resistance (*R*) (Ω) | Uncertainty (K) |
|---|---|---|
| -38.8309 | 84.71680 | 0.020 |
| 0.0102 | 100.00717 | 0.020 |
| 29.7646 | 111.59964 | 0.020 |
| 156.5979 | 159.83310 | 0.020 |
| 231.9292 | 187.59997 | 0.020 |
| 300.0155 | 212.13557 | 0.030 |
| 449.9971 | 264.32952 | 0.030 |

NOTE:

The calibration results were fitted by using least square method to obtain the unique constants, *A*, *B* and *C*. The temperature-resistance relationships can be calculated using the interpolation function below:

For the range 0 °C to 450 °C

$R_t \quad = \quad R_0\,(1 + At + Bt^2)$

and for the range - 38 °C to 0 °C

$R_t \quad = \quad R_0\,(1 + At + Bt^2 + C(t-100)t^3)$

Where;

$R_t$ = resistance (in ohms) at temperature, t (°C)

$R_0$ = 100.006 612 Ω

$A$ = 0.003 909 63 °C$^{-1}$

$B$ = -5.739 697 $\bullet 10^{-7}$ °C$^{-2}$

$C$ = -2.561 645 $\bullet$ $10^{-11}$ °C$^{-4}$

The estimated temperature measurement uncertainty when using this equation for whole range is 0.05K

And the temperature coefficient $\alpha$ was calculated from $\alpha = \frac{R_{100} - R_0}{100 R_0}$ = 0.003 852 23 °C$^{-1}$

**APPENDIX E** **Calculate the constants values**

From Interpolation Equations for Industrial Platinum Resistance Thermometers, the constants: Calandar Vandusen-Coefficient [*A, B, C*] can be calculate as example below.

**For Example;**
From reference table ASTM-E1137 the resistance values corresponding to temperature are list below:

| **t ( °C)** | **$R_t$ (Ω)** |
|---|---|
| 600 | 313.71 |
| 400 | 247.09 |
| 200 | 175.86 |
| 0 | 100.00 |
| -200 | 18.52 |

For temperature range [ $0°C \leq t < 650°C$ ], the interpolation equation is:

$$R_t = R_0 ( 1 + At + Bt^2 )$$
$$( R_t / R_0 ) - 1 = At + Bt^2$$

at 200 °*C* $0.758\,6 = 200A + 40\,000B$ (1)

at 400 °*C* $1.470\,9 = 400A + 160\,000B$ (2)

at 600 °*C* $2.137\,1 = 600A + 360\,000B$ (3)

(1)+ (2) $2.229\,5 = 600A + 200\,000B$ (4)

(3) - (4) $-0.092\,4 = 160.000B$

$$B = ( -0.0924) \div 160{,}000$$
$$= -0.000\,000\,577\,537\,314$$
$$= -5.775 \times 10^{-7}$$

from Eq.(4) calculate constant *A*

$$2.2295 = 600A + 200{,}000 \times (-0.000000577537314)$$
$$600A = 2.2295 + 0.115\,507\,462$$
$$A = 2.345\,007\,462 \div 600$$
$$= 0.00390834577$$
$$= 3.9083 \times 10^{-3}$$

For temperature range [$-200\ °C \leq t < 0°C$], the interpolation equation is:

$$R_t = R_0 ( 1+At+Bt^2+C(t-100)t^3 )$$
$$( R_t / R_0 )-1 = At+Bt^2+C(t-100)t^3$$

at: [-200 °C]

$$(18.52/100)-1 = 0.003\,908\,345\,77(-200) + (-0.000\,000\,577\,537\,314)(-200)^2 + C[(-200)-100](-200)^3$$

$$-0.814\,8 = -0.781\,669\,154 - 0.023\,101\,492 + 2{,}400{,}000{,}000\,C$$

$$C = -0.814\,8 + 0.781\,669\,154 + 0.023101492$$

$$= -0.010\,029\,354 \div 2{,}400{,}000{,}000$$

$$= -4.178\,897\,5 \times 10^{-12}$$

$$(18.52/100)-1 = 0.003\,908\,3(-200) + (-0.000\,000\,577\,5)(-200)^2 + C[(-200)-100](-200)^3$$

$$C = -0.814\,8 + 0.781\,66 + 0.023\,1$$

$$= -0.010\,04 \div 2{,}400{,}000{,}000$$

$$= -4.183\,3 \times 10^{-12}$$

The calculation functions are also available in some commercial software such as Microsoft Excel.